C2.1 Chemical signalling

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-c2-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 450–466 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Signals and receptors

C2.1.1 Receptors and ligands

pp. 450–451

A **ligand** is a signalling chemical; a **receptor** is a protein with a binding site that fits one ligand. Binding changes the receptor's shape and starts a response, so only cells with the right receptor respond.

C2.1.2 Quorum sensing

pp. 452–453

Bacteria release an **autoinducer**. At high cell density it passes a threshold, binds receptors and switches on genes in all the cells. *Vibrio fischeri* in the bobtail squid's light organ switches on luciferase genes and glows only when the population is dense.

C2.1.3 Kinds of signalling chemical

pp. 453–454

Hormones travel in the blood; **neurotransmitters** cross synapses; **cytokines** act on nearby cells, often in immunity; **calcium ions** act inside cells, for example triggering muscle contraction.

C2.1.4 Chemical diversity

p. 454

Hormones may be amines (adrenaline), proteins (insulin) or steroids (oestradiol). Neurotransmitters include amino acids, peptides, amines and the gas nitric oxide. Each receptor needs a distinct shape, and different chemistry suits different jobs.

C2.1.5 Local and distant effects

p. 455

Neurotransmitters act across a gap of about 20 nm within milliseconds; hormones travel round the body in the blood and act on every cell with the right receptor, often for hours.

C2.1.6 Transmembrane and intracellular receptors

pp. 456–457

Hydrophilic ligands bind **transmembrane receptors**, which have hydrophobic amino acids where they cross the membrane. Hydrophobic ligands such as steroids diffuse into the cell and bind **intracellular receptors**, which are mostly hydrophilic.

Inside the cell

C2.1.7 Signal transduction

p. 458

Binding of a ligand starts a **signal transduction pathway**: a sequence of changes inside the cell that passes on and amplifies the signal, ending in a response.

C2.1.8 The acetylcholine receptor

pp. 459–460

Acetylcholine binds a receptor that is also an ion channel. The channel opens, positive ions (mainly Na^+) diffuse in and the membrane depolarizes, which may start an impulse.

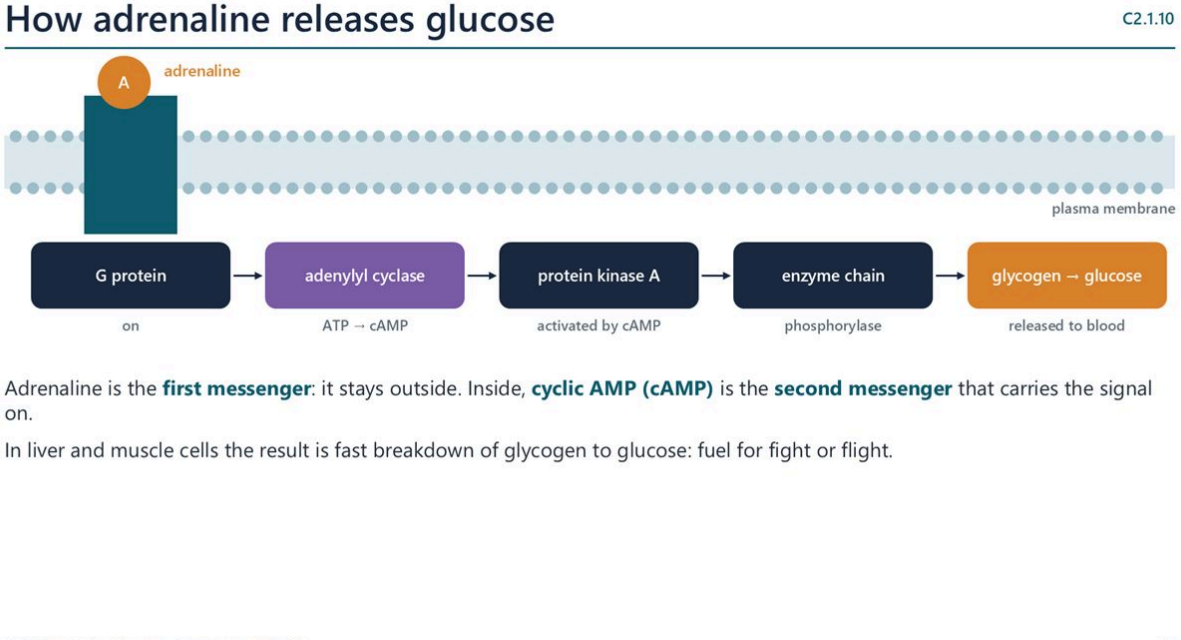
C2.1.9 G protein-coupled receptors

pp. 460–461

A ligand binding outside changes the receptor's shape; the G protein inside swaps GDP for GTP and activates an enzyme or channel. Humans have hundreds of these receptors.

C2.1.10 Adrenaline

pp. 461–462



Adrenaline binds a G protein-coupled receptor; the G protein activates adenylyl cyclase, which makes **cAMP**, the second messenger. cAMP activates protein kinase A and a cascade that breaks down glycogen, amplifying the signal many times.

C2.1.11 Insulin and tyrosine kinase

pp. 462–463

Insulin binds a receptor that phosphorylates **tyrosine** inside the cell. A chain of reactions moves vesicles carrying **GLUT4** transporters to the membrane, so more glucose enters the cell.

C2.1.12 Intracellular receptors

p. 464

Oestradiol, progesterone and testosterone diffuse into cells and bind receptors there. The activated receptor binds specific DNA sequences and switches on transcription of target genes.

C2.1.13 Oestradiol and progesterone

p. 465

Oestradiol acts on cells in the hypothalamus that secrete GnRH, changing its release. Progesterone acts on cells of the endometrium, keeping the lining thick and ready for an embryo.

C2.1.14 Feedback

pp. 465–466

Negative feedback reduces the signal and keeps a variable stable (insulin lowering blood glucose). **Positive feedback** increases it and drives a rapid change (an autoinducer switching on its own production).

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